

ROVER LAB, DOSSIER (LHS 1140 b mission)

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Planet

Name: LHS 1140 b

Planet type: Dense, water-rich rocky super-Earth

Surface temperature: ≈ -40 °C equilibrium estimate (range -45 to -38 °C)

Surface pressure: Not directly measured; JWST data hint at (not yet confirmed) a thin nitrogen- or CO₂-dominated atmosphere

Gravity: ≈ 1.9 g

Atmosphere: A hydrogen-rich atmosphere has been firmly ruled out (JWST, $>10\sigma$); tentative, unconfirmed evidence points toward a thin N₂- or CO₂-dominated atmosphere

Mission Briefing

LHS 1140 b is a dense, cold rocky super-Earth thought to hold a substantial fraction of water ice and rock, orbiting a quiet, calm red dwarf. The rover 'Frostline-1' has landed near a shallow, frozen basin at the planet's mild sub-stellar region, where a thin crust of pale, rust-tinted salts has been detected. Orbital data suggest this could be an ancient brine, slowly concentrated over eons as the surface froze and thawed at the edges of a retreating ocean. Mission control hopes it contains enough dissolved iron to be worth mining for the base's radiation shielding.

Rover

Starting energy: 100

Site visit cost: 10

Experiment cost: 15

Overview Image

Site: Pale, rust-tinted crust at the basin's edge

Description: A thin, frost-rimmed crust with a faint rusty tint, cracking underfoot at the edge of a shallow frozen basin.

Identity: Iron(III) chloride solution (real sample, suggested ≈ 0.1 M), medium-low concentration. REAL SAMPLE GROUP.

Known Properties

Colour: Pale yellow-orange, lighter than more concentrated samples

Odour: Very faint

Solubility: Fully miscible with water

Experiment Results

Measure pH: Mildly acidic, pH around 2.3–2.6

Measure density: 1.06 g/mL

Test electrical conductivity: Moderate

Add a magnet: No attraction

Add potassium thiocyanate solution: Turns red, though less intensely than a more concentrated sample

Add sodium hydroxide solution: Forms a light reddish-brown precipitate

Heat a small sample gently: No strong visible change

Experiments

Measure pH

Measure density

Test electrical conductivity

Add a magnet

Add potassium thiocyanate solution

Add sodium hydroxide solution

Heat a small sample gently

AI Behaviour

Reveal remaining candidates after narrowing: No

Socratic questions: Yes

Difficulty: Standard (15-16 years old)

Teacher Notes

REAL SAMPLE GROUP, assign real FeCl₃, medium-low concentration (suggested ≈ 0.1 M). LHS 1140b is cold and likely water/ice-rich, so it's framed as more dilute than the hotter, drier planets, continuing the evaporation-concentration logic used across the other real groups.

Real-science notes: LHS 1140b is a genuine, actively studied rocky exoplanet, ~ 48.8 light-years away, discovered in 2017 (MEarth Project), orbiting its red dwarf every 24.7 days. It is widely considered one of the single best real targets for eventually detecting a genuine rocky-exoplanet atmosphere, precisely because its high gravity and low stellar irradiation make atmospheric loss less likely than for most other candidates. Its measured density (recomputed in 2024, Cadieux et al.) is too high for a gas-rich mini-Neptune and instead points to a rocky planet with a substantial water-rich layer (roughly 9–19% water by mass), a genuine 'water world' candidate. JWST has firmly ruled out a hydrogen-rich atmosphere and found tentative (unconfirmed) hints consistent with a thin nitrogen- or CO₂-dominated atmosphere, the lead author has said more telescope time is needed to confirm this. No iron or chloride detection of any kind exists for this planet; as with the others, the mission/deposit is an invented framing.

Sources: Cadieux et al. 2024, ApJL (iopscience.iop.org/article/10.3847/2041-8213/ad5204); arXiv (arxiv.org/pdf/2406.15136).

Cross-subject link: Biology selective pressure for this planet, very dim red-dwarf starlight combined with unusually high surface gravity (~ 1.9 g).